BATTERY MANAGEMENT SYSTEM

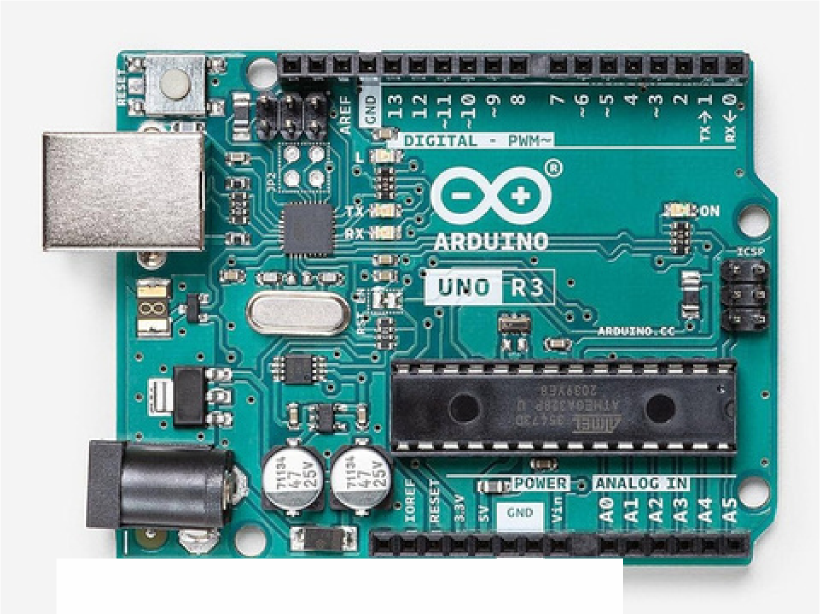
PROBLEM STATEMENT

- Lithium-Ion Instability: Lithium-ion cells have narrow safety margins. Overcharging them or exposing them to excessive heat can lead to catastrophic thermal runaway, fires, or explosions.
- Irreversible Cell Degradation: Drawing too much power or allowing the pack voltage to drop below critical thresholds permanently damages the internal chemistry, destroying the battery's capacity and lifespan.
- Blind System Operation: Microcontrollers and external loads cannot natively read raw battery conditions, leaving the host system completely unaware of imminent power failures or critical hazards.
- The Engineering Objective: To design and implement a low-cost, automated Battery Management System (BMS) that continuously monitors physical cell parameters — including voltage, current, and temperature — and executes rapid load disconnection to protect the battery pack during fault conditions.

COMPONENTS REQUIRED

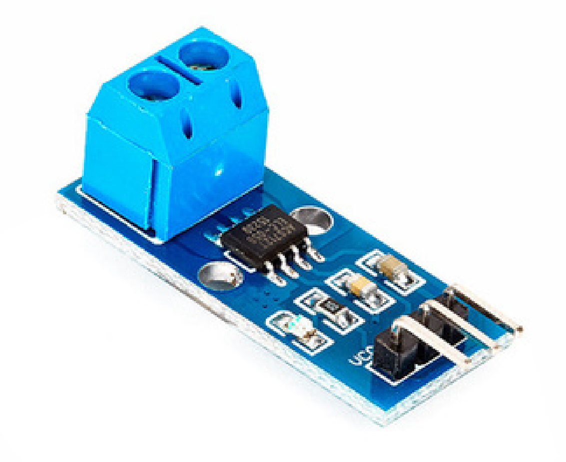
Arduino Uno R3

- The Brain: Central microprocessing hub for the whole BMS.
- ADC Reader: Converts 0V-5V analog signals into digital data .
- Pins: Inputs data via pins A0-A2; outputs trip signals via pins D2-D3.



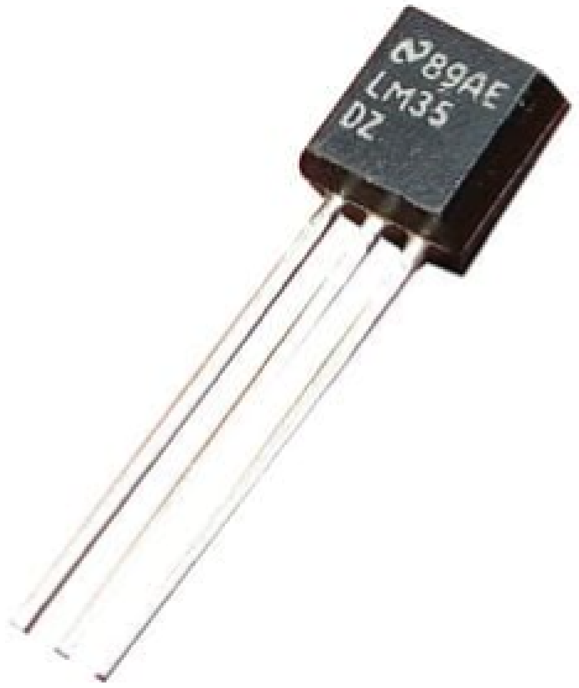
ACS712 Current Module

- Amperage Tracker: Measures current flowing in or out of the battery pack.
- Hall Effect: Translates magnetic fields from current into readable voltage.
- 0A Baseline: Outputs exactly 2.5V when no current is moving.



LM35 Temperature Sensor

- Heat Sentinel: Placed directly near cells to continuously monitor battery temperature and catch overheating in real time
- Direct Voltage Output: Outputs exactly 10mV per °C — no resistance conversion needed unlike NTC thermistor
- Simple Connection: Only 3 pins needed — VCC, OUT and GND — no pull up resistor required
- Accurate Reading: Gives precise temperature readings with ±0.5°C accuracy directly to Arduino A2 in range 0°C to 100°C



COMPONENTS REQUIRED

5V Single-Channel Relay

- Circuit Breaker: Acts as a heavy-duty isolation switch during a fault condition, physically disconnecting the load from the battery pack.
- Electromagnetic Operation: Low-power Arduino digital pin (D2) trips an internal coil to snap the contacts open, requiring minimal current to control high-power circuits.
- Load Protection: When a fault is detected — overcurrent, overvoltage, undervoltage or overtemperature — the relay immediately disconnects the DC motor load to protect the battery pack.

18650 Lithium-Ion Battery Pack

- The Powerhouse: Main energy source being monitored and protected.
- Cell Configuration: 2S configuration (2 cells in series) giving an 8.4V}peak.
- Voltage Limits: Operates strictly between 6.0V (empty) and 8.4V (full).
- The Risk: Highly sensitive to over-discharge, over-charge, and fire risks (thermal runaway).

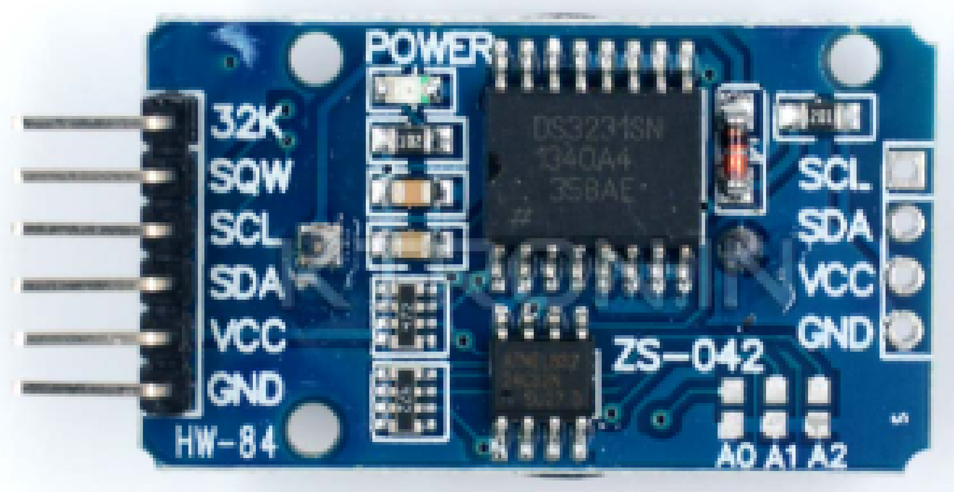
Voltage Divider Network

- Voltage Scaler: Drops the raw battery pack voltage to a safe readable level for the Arduino analog pin.
- Resistor Pair: Uses two series resistors — 30kΩ and 10kΩ — to step down the maximum battery voltage of 8.4V to approximately 2.1V at the midpoint.
- Protection: Keeps the analog input voltage safely under 5V to prevent damage to the Arduino's ADC pin, while maintaining accurate proportional voltage readings.

COMPONENTS REQUIRED

DS3231 RTC Module

- Precision Timekeeper: Maintains exact real time using built in crystal oscillator — far more accurate than NE555 timer which drifts with temperature and component tolerance
- I2C Communication: Connects to Arduino via only 2 wires — SDA on A4 and SCL on A5 — for two way data exchange
- Alarm Function: Arduino sets precise 3 minute recovery alarm inside RTC via I2C before tripping relay during fault
- Always Accurate: After exactly 3 minutes RTC notifies Arduino to restore power — no hardware resistors or capacitors needed to set time, just change a number in code

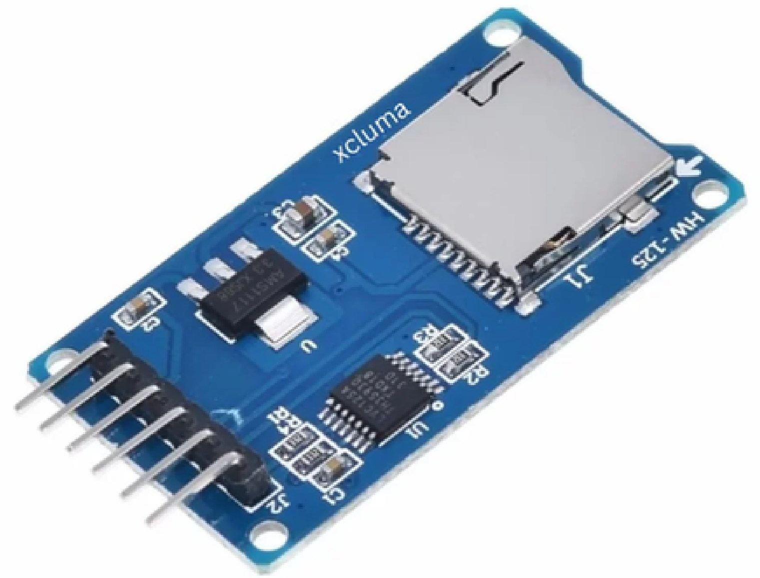


AMS1117 5V Voltage Regulator

- Stable Power Supply: Converts fluctuating battery voltage 7.4V - 8.4V into clean stable 5V to keep Arduino always powered
- Direct Battery Connection: Connected directly to battery positive — completely independent from relay switching — ensuring Arduino never loses power during fault or recovery
- Protection: Shields Arduino from voltage spikes and fluctuations that could cause crashes or damage
- Always Alive: Since Arduino is continuously powered via AMS1117, it can monitor sensors, count recovery time and control relay entirely on its own without needing external recovery components

SD Card Module

- Data Logger: Records every fault event with exact timestamp, fault type, voltage, current and temperature
- SPI Communication: Uses 4 wire SPI protocol on Arduino pins D11, D12, D13 and D10
- Permanent Storage: Saves data to microSD in CSV format readable on any computer



WORKING

Stage 1: Raw Data Collection

- Continuous Sampling: Sensors directly track battery pack's physical state in real time
- Voltage Scaling: Voltage divider steps down battery voltage 6.0V - 8.4V into safe 0V - 2.1V window for Arduino
- Signal Routing: Scaled voltage from divider, current readings from ACS712, and temperature from LM35 sent to Arduino pins A0, A1, and A2
- Power Regulation: AMS1117 5V regulator converts battery voltage directly to stable 5V keeping Arduino always alive

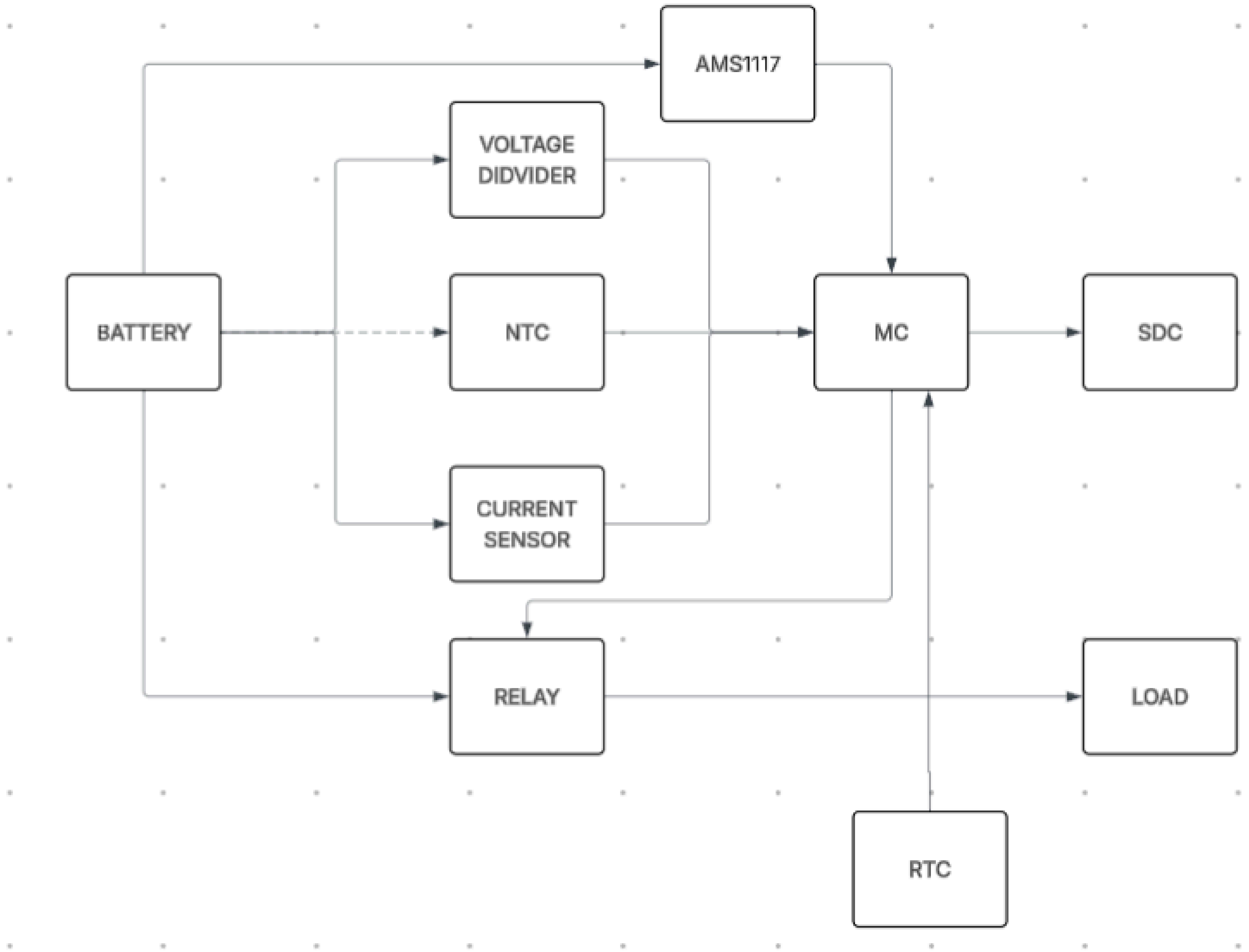
Stage 2: Logic Evaluation

- ADC Conversion: Arduino Uno converts incoming analog signals into digital values 0 to 1023
- Unit Translation: Internal code applies math formulas to convert digital numbers into Volts, Amps and degrees Celsius
- Threshold Checking: Code runs continuous loop comparing live values against built in safety limits
- RTC Timekeeping: DS3231 RTC communicates with Arduino via I2C on pins A4 and A5 maintaining precise time tracking

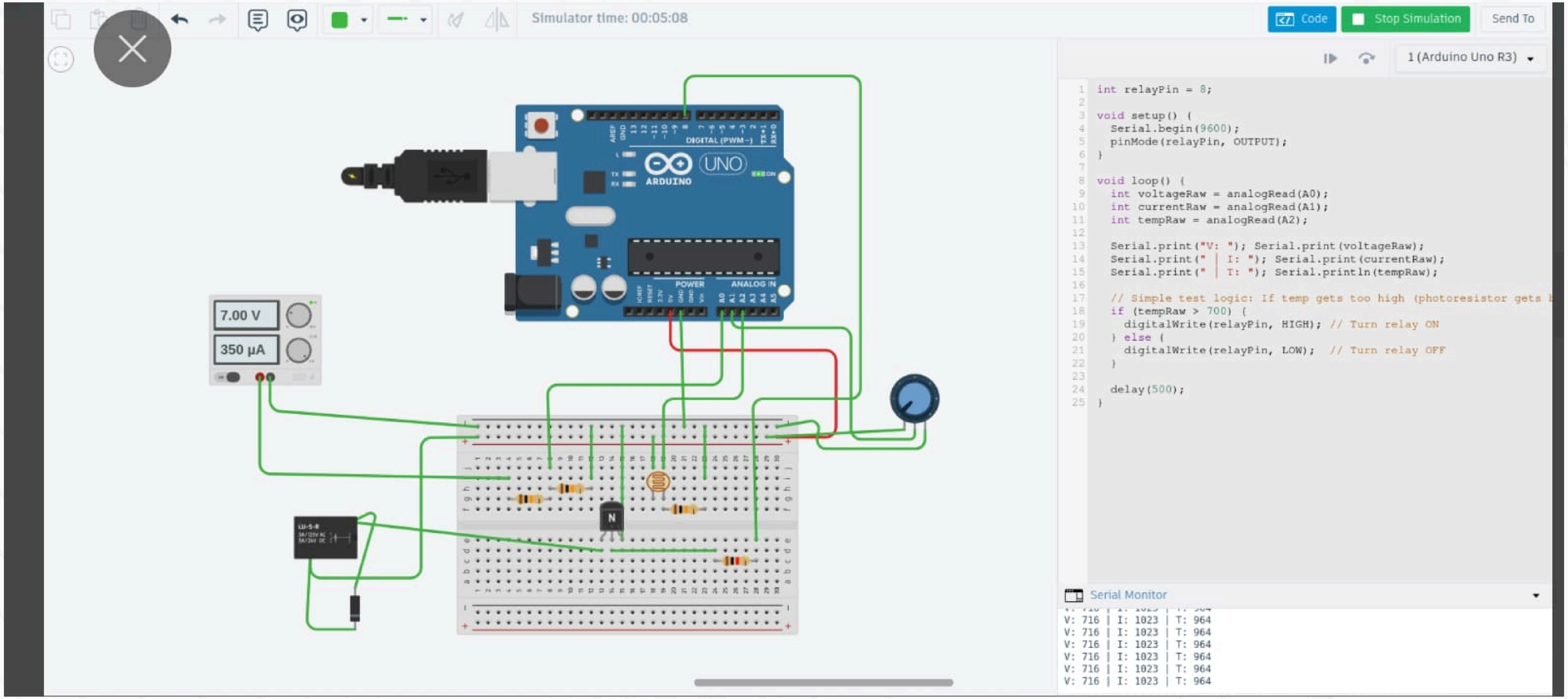
Stage 3: Fault Execution

- Over Voltage / Over Heat Fault: If voltage exceeds 8.4V or temperature crosses 45°C Arduino sends HIGH signal to D2 tripping relay and disconnecting motor load
- Under Voltage / Over Current Fault: If voltage drops below 6.0V or current exceeds 5A Arduino triggers relay to disconnect load preventing further cell damage
- Fault Logging: Arduino sets precise 3 minute recovery alarm in RTC via I2C before cutting power
- Auto Recovery: After exactly 3 minutes RTC notifies Arduino via I2C, Arduino sets D2 LOW, relay closes, motor restarts and sensor check begins again
- System Status: If all variables within safe limits relay remains closed and motor operates normally ✅

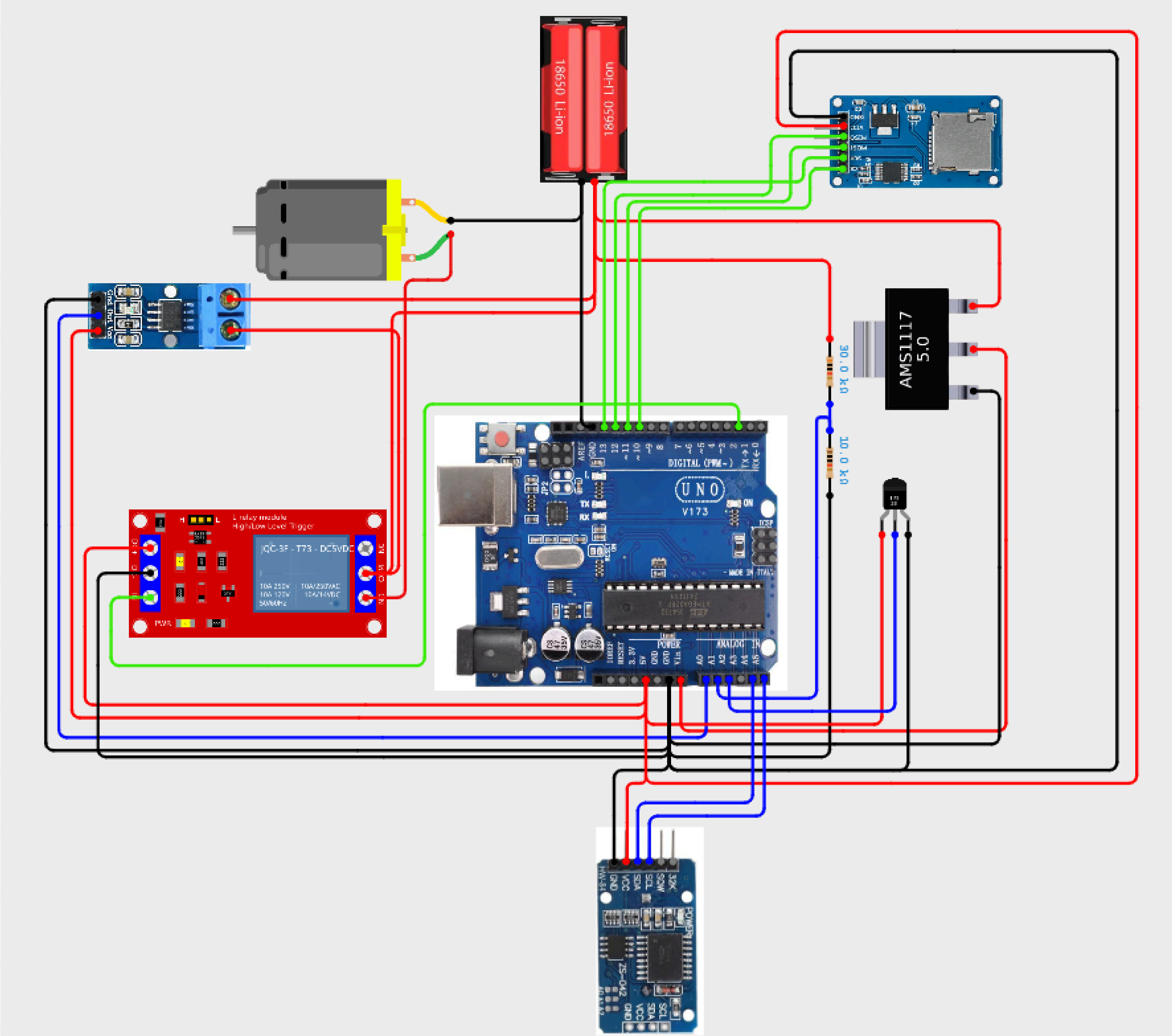
BLOCK DIAGRAM



IMPLEMENTATION DIAGRAM



CIRCUIT DIAGRAM



BMS – Bill of Materials							
SL	**Component**	**Specification**	**Legs / Pins**	**Qty**	**Unit Price (Rs.)**	**Subtotal (Rs.)**	**Supplier (Best Price)**
1	18650 Li-ion Cell	3.7V nominal / 4.2V max per cell Pack: 7.4V (8.4V full)	2 terminals	2	Rs.200 - Rs.350	Rs.400 - Rs.700	robu
2	18650 Battery Holder	Holds 2x 18650 in series	2 wire leads	1	Rs.21 - Rs.65	Rs.21 - Rs.65	Tomsonelectronics
3	Arduino UNO R3	ATmega328P @ 16MHz Vin: 5–12V	32 pins	1	Rs.270 - Rs.550	Rs.270 - Rs.550	Robocraze
4	AMS1117	Input: up to 18V · Output: 5V fixed	3 legs	1	Rs.10 - Rs.25	Rs.10 - Rs.25	Robu
5	LM35 Temperature Sensor	Range: -50°C to +150°C	3 legs	1	Rs.40 - Rs.80	Rs.40 - Rs.80	Robu
6	ACS712 Current Sensor	Range: ±20A	3 pins (header) 2 pins (power path)	1	Rs.115 - Rs.180	Rs.115 - Rs.180	Robocraze
7	DS3231 RTC Module	VCC: 2.3–5.5V	6 pins	1	Rs.150 - Rs.350	Rs.150 - Rs.350	Robu
8	SD Card Module	3.3V / 5V	6 pins	1	Rs.60 - Rs.120	Rs.60 - Rs.120	Robocraze
9	5V Relay Module	Coil: 5V DC	5 pins	1	Rs.60 - Rs.100	Rs.60 - Rs.100	Robocraze
10	Resistor 10kΩ	10kΩ ±5%	2 legs	1	Rs.1 - Rs.2	Rs.1 - Rs.2	All similar
11	Resistor 30kΩ	30kΩ ±5%	2 legs	1	Rs.1 - Rs.2	Rs.1 - Rs.2	All similar
12	Resistor 1kΩ	1kΩ ±5%	2 legs	1	Rs.1 - Rs.2	Rs.1 - Rs.2	All similar
Components Subtotal						Rs.1,129 - Rs.2,176	—
PCB						Rs.80 - Rs.150	—
GRAND TOTAL						**Rs.1,169 - Rs.2,226**	—

ADVANTAGES AND DISADVANTAGES

Advantages

- Low Cost: Uses readily available components — Arduino, relay, ACS712 — keeping the total build cost minimal.
- Real-Time Protection: Continuously monitors voltage, current and temperature simultaneously, reacting instantly to faults.
- Modular Design: Each sensor operates independently, making it easy to troubleshoot or replace individual components.
- Open Source: Arduino platform allows full code customization and easy future upgrades.

Disadvantages

- No Cell Balancing: Cannot manage individual cell voltages in a multi-cell pack, leading to gradual cell mismatch over time.
- Load Only Protection: Current design only disconnects the load during a fault — charger isolation requires additional hardware.
- Analog Noise: Analog sensor readings are susceptible to electrical noise which can cause false fault triggers.
- No SoC Estimation: Cannot calculate precise battery percentage — only monitors raw voltage as an approximation.

FUTURE SCOPE

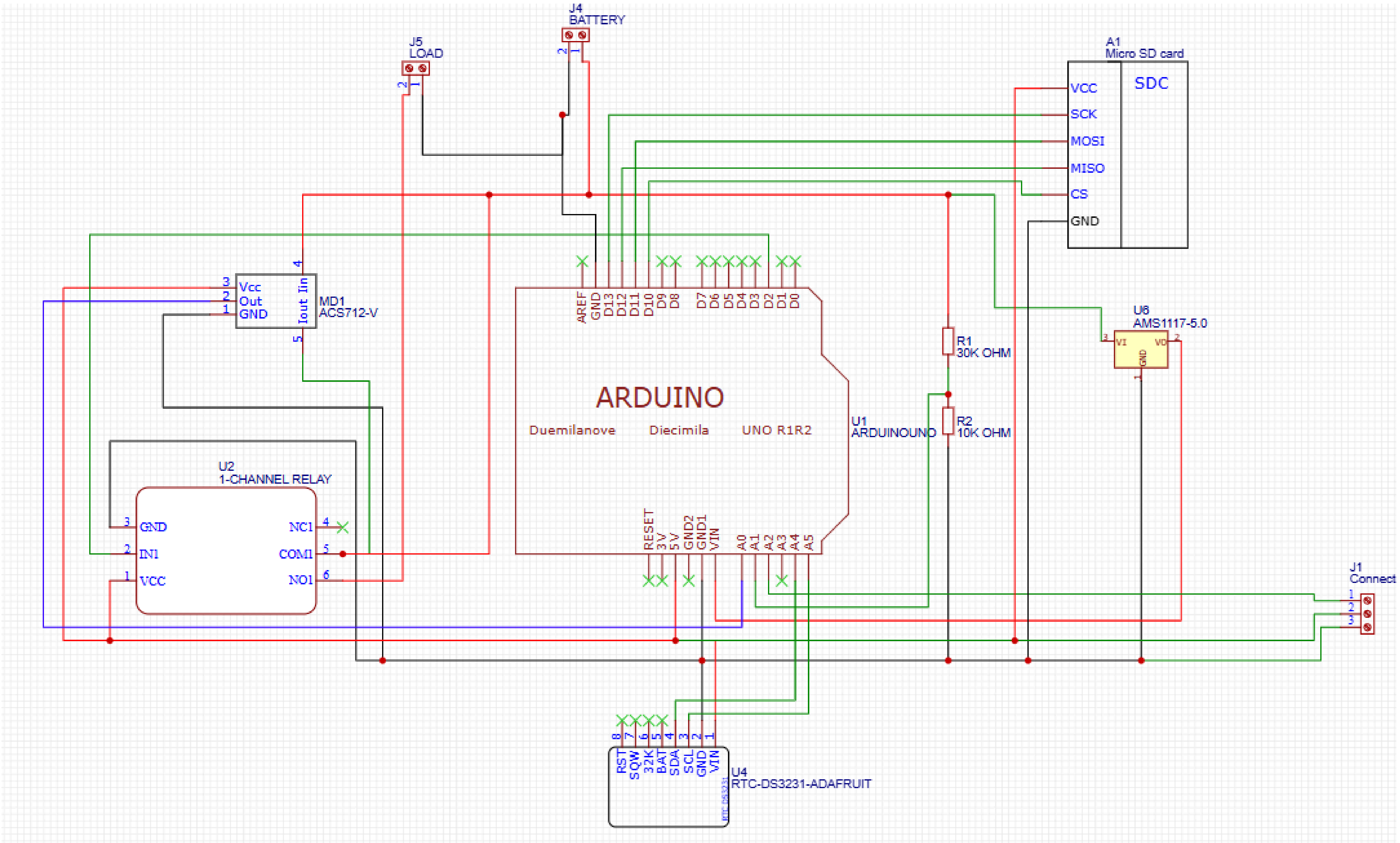
1. Digital Protocols
    - CAN-bus / I2C: Upgrade from simple analog wires to a high-speed digital data bus.
    - Benefit: Eliminates electrical noise and allows communication with industrial microcontrollers.

2. Active Cell Balancing
    - Individual Monitoring: Track and manage the voltage of each individual 18650 cell.
    - Energy Transfer: Actively move power from stronger cells to weaker cells to maximize the pack's lifespan.

3. IoT & Wireless Monitoring
    - ESP32 Upgrade: Swap the Arduino Uno for an ESP32 microchip with onboard Wi-Fi and Bluetooth.
    - Cloud Dashboard: Stream live battery data directly to a mobile app or remote web server.

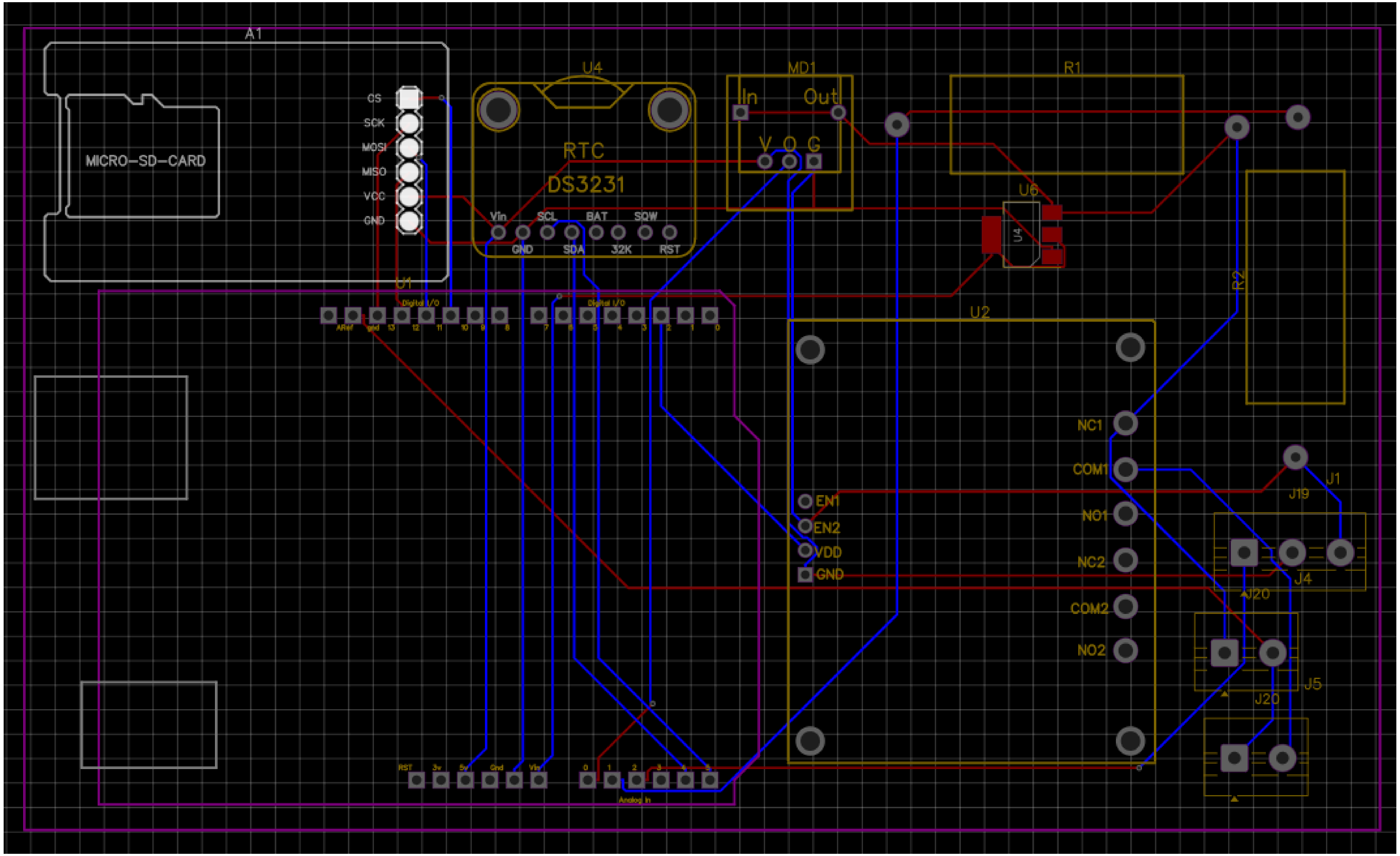
4. Smart State-of-Charge (SoC)
    - Coulomb Counting: Track the exact quantity of electrical charge entering and leaving the pack.
    - Accuracy: Replaces basic voltage-guessing with a precise, smartphone-like battery percentage.

SCHEMATIC



J4
BATTERY
J5
LOAD
A1
Micro SD card
SDC
VCC
SCK
MOSI
MISO
CS
GND
MD1
ACS712-V
Vcc
Out
GND
Iin
Iout
U6
AMS1117-5.0
VI
VO
GND
R1
30K OHM
R2
10K OHM
ARDUINO
Duemilanove
Diecimila
UNO R1R2
U1
ARDUINOUNO
AREF
GND
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1
D0
RESET
3V
5V
GND2
GND1
VIN
A0
A1
A2
A3
A4
A5
U2
1-CHANNEL RELAY
GND
IN1
VCC
NC1
COM1
NO1
J1
Connect
U4
RTC-DS3231-ADAFRUIT
RST
SQW
32K
BAT
SDA
SCL
GND
VIN

PCB


A1
MICRO-SD-CARD
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GND
SDA
32K
RST
MD1
In
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V O G
R1
U6
R2
U1
U2
NC1
COM1
NO1
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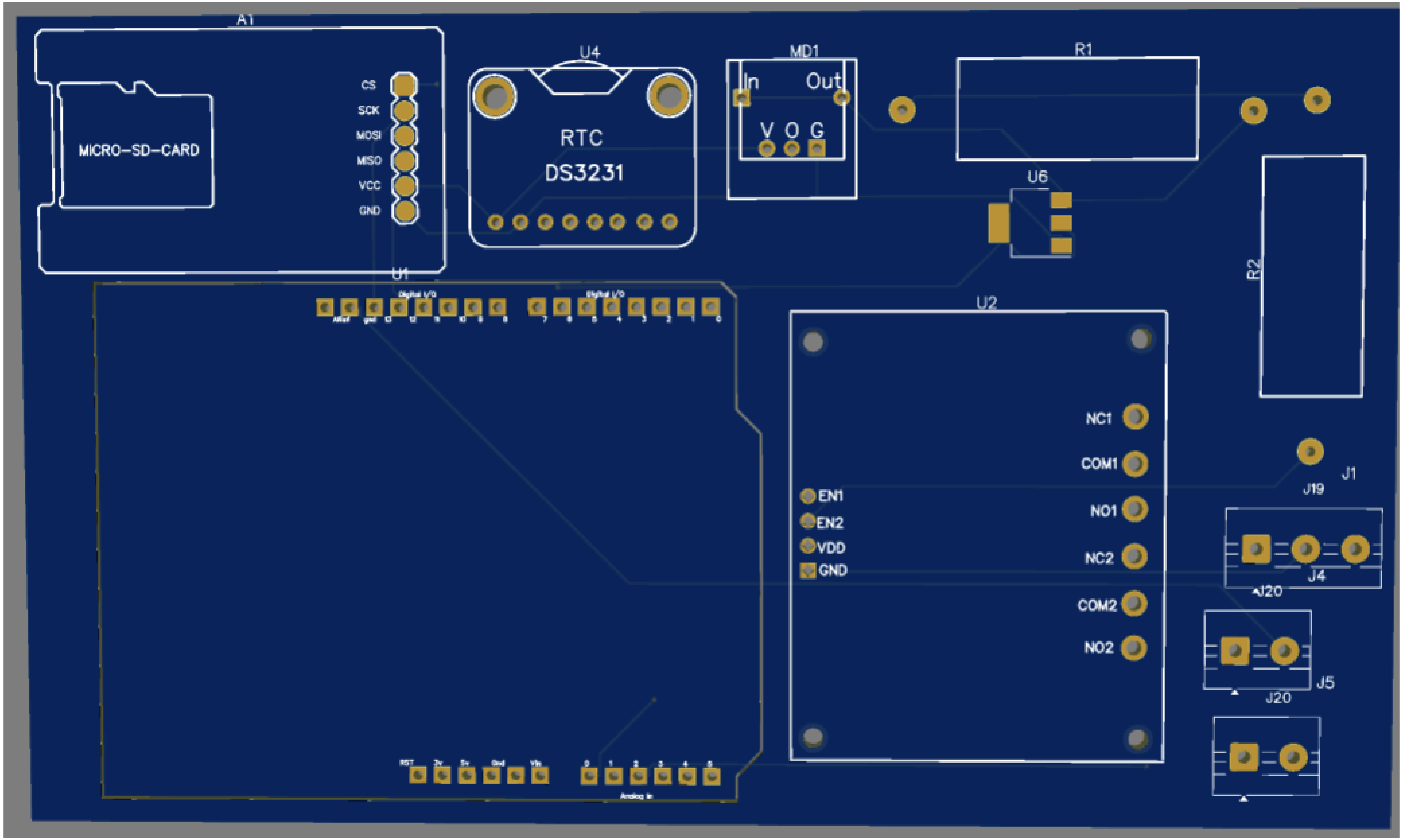
COM2
NO2
EN1
EN2
VDD
GND
J1
J19
J4
J20
J5

3D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